5V DC To 48V DC CONVERTER For Phantom Power Supplies

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This circuit presents a simple and low-cost DC-to-DC converter for phantom power supplies used in mixing consoles, microphone preamplifiers, telephone systems and similar equipment. There are many application circuits that require 24V and 48V phantom power supplies, but these DC power supplies are costly and not easily available. 5V DC power supplies are easily available at low cost. This circuit presents a solution for converting 5V DC to 48V DC using a popular MC34063A IC.

Circuit and working

The circuit diagram of the 5V DC to 48V DC converter is shown in

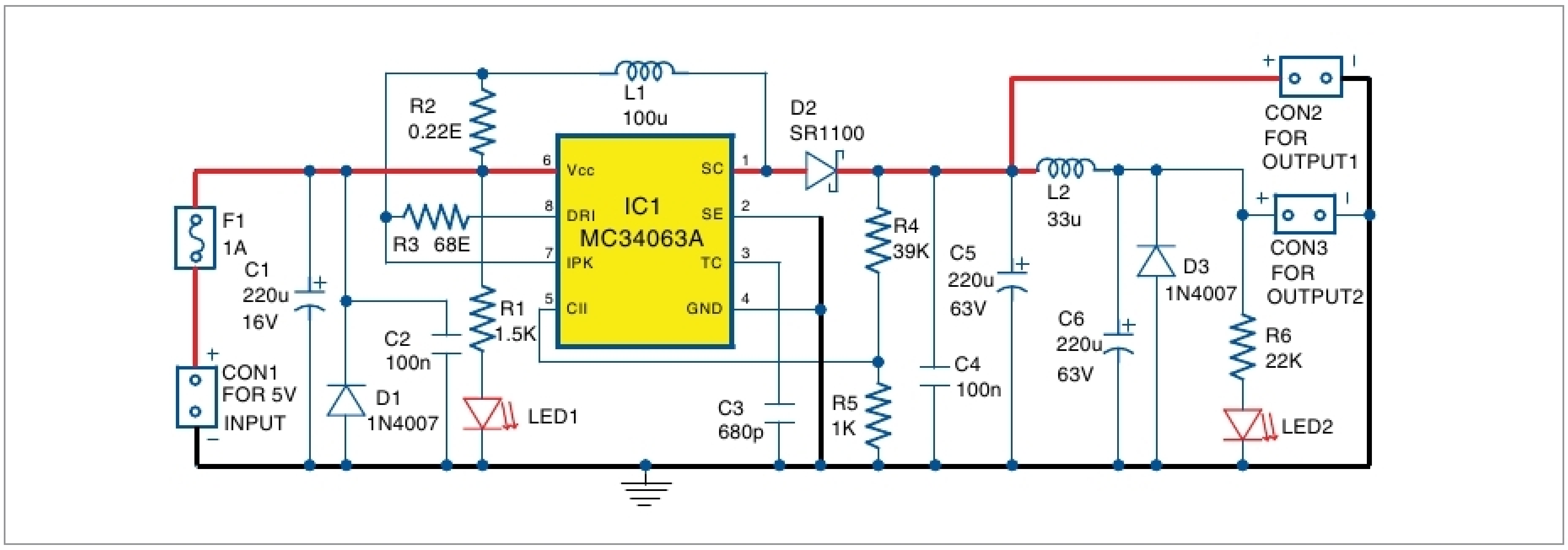

Fig. 1: Circuit diagram of 5V DC to 48V DC converter

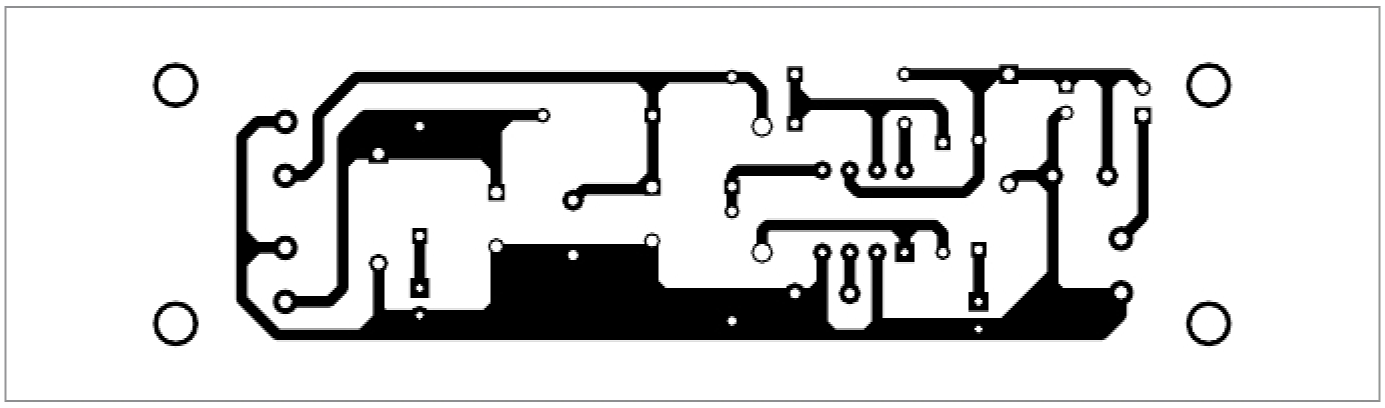

Fig. 2: Actual-size PCB layout of DC-to-DC converter

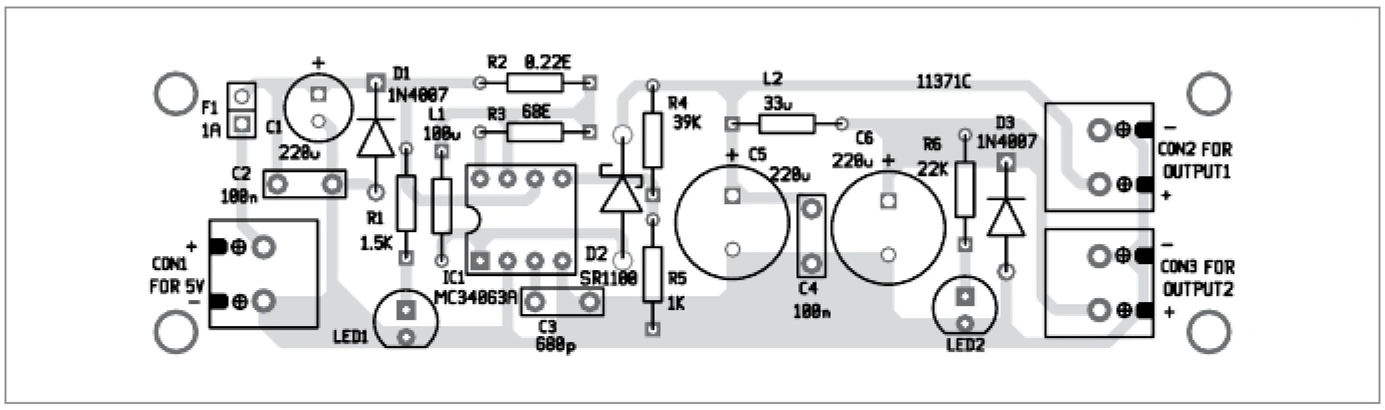

Fig. 3: Components layout for the PCB

Fig. 1. It is built around boost/buck/inverting switching regulator MC34063A (IC1), two rectifier diodes 1N4007 (D1 and D3), Schottky diode SR1100 (D2), two 5mm LEDs (LED1 and LED2), two inductors (L1 and L2) and a few other components.

Input power supply ranging from 5V to 7V can be applied to connector CON1. Fuse F1 protects the circuit from any continuous over-current input. Diode D1 protects the circuit from reverse input voltage. Resistor R2 limits the maximum switching current to around 1.5A, as per relationship:

Ipeakmax = 0.3V/R2

Resistor R3 limits the